PlantCare AI

Updated Source Code and Project Explanation

A responsive college-project prototype for plant leaf image preview and plant-care guidance.

Important: The current code demonstrates a fixed sample result. It does not include a trained AI model and does not actually diagnose the uploaded image.

Project Explanation

PLANTCARE AI — UPDATED PROJECT EXPLANATION

1. Project Overview
PlantCare AI is a responsive front-end prototype for a plant disease detection and plant-care guidance system. It is designed as a college AI Immersion project. Users can select a plant leaf image, preview it, and view an illustrative sample disease result with care suggestions.

2. Problem Statement
Plant diseases can affect crop quality and yield. Identifying symptoms early may be difficult when farmers must inspect many plants manually. A simple digital interface can help demonstrate how image-based crop monitoring could be presented.

3. Proposed Solution
The website provides Home, AI Detection, Solution, Prevention, and About sections. It lets a user choose a plant image or open the camera/file chooser on supported devices. After the user presses Run Demo Detection, the page displays a sample Tomato Early Blight result and five general care steps.

4. Important Current Limitation
This version does NOT contain a trained AI model and does NOT analyse the leaf pixels to identify a disease. Tomato Early Blight and the 94% confidence value are fixed demonstration data. The confidence value is illustrative, not a measured prediction. Do not present this prototype as a real AI diagnosis system.

5. Main Features
- Responsive green agriculture-themed interface for mobile and desktop.
- Navigation between sections without reloading the page.
- Plant leaf image selection and image preview.
- Camera/file chooser support where the browser and device allow it.
- Alert when detection is requested without an image.
- Fixed sample result with symptoms and severity.
- Five educational plant-care steps.
- Prevention, watering, nutrition, and expert-support guidance.
- Dashboard counters for total demo scans and sample disease results.

6. Technologies Used
HTML: defines the structure and sections of the website.
CSS: controls colors, layout, cards, spacing, and mobile responsiveness.
JavaScript: switches sections, validates image selection, displays previews, shows the sample result, and updates counters.

7. How to Run
1. Save the supplied source as index.html.
2. Open index.html in a modern web browser.
3. Open AI Detection.
4. Select a plant leaf image or use the camera/file chooser.
5. Press Run Demo Detection to view the fixed sample result.
6. Navigate to Solution or Prevention to read the guidance.

8. Expected Outcome
The user can navigate the website, select a leaf image, view the image preview, and see a sample result and plant-care suggestions. Dashboard values increase when the demo detection button is pressed.

9. Future Enhancements
- Train or integrate a plant disease image classification model using TensorFlow/Keras.
- Connect a backend API for real model inference.

- Add verified crop-specific recommendations and local-language support.
- Add voice input/output and secure scan-history storage.
- Test the model on varied real-world images and report validated performance metrics.

10. Safety Note

This project is for educational demonstration only. Plant symptoms can look similar across different causes. Confirm serious, unfamiliar, or spreading plant problems with a qualified agriculture professional before taking treatment decisions.

Complete Source Code (index.html)

Save this code as **index.html** and open it in a modern browser.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="PlantCare AI is a student project prototype for plant leaf image
    preview and plant-care guidance.">
  <title>PlantCare AI | Smart Plant Care</title>
  <style>
    :root {
      --green: #087a45;
      --dark: #063b28;
      --light: #f1fff5;
      --mint: #e8f8ee;
      --text: #17352a;
      --muted: #5c7166;
      --white: #ffffff;
      --border: #cfe8d8;
      --warning: #fff4d5;
    }
    * { box-sizing: border-box; margin: 0; padding: 0; }
    body {
      font-family: Arial, Helvetica, sans-serif;
      background: var(--light);
      color: var(--text);
      line-height: 1.6;
    }
    header {
      background: var(--green);
      color: var(--white);
      padding: 16px;
      text-align: center;
      position: sticky;
      top: 0;
      z-index: 5;
      box-shadow: 0 3px 12px #0002;
    }
    header h1 { font-size: 1.6rem; }
    nav { display: flex; flex-wrap: wrap; justify-content: center; gap: 8px; margin-top: 12px; }
    nav button, .btn {
      border: 0;
      border-radius: 10px;
      padding: 10px 14px;
      font-weight: bold;
      cursor: pointer;
      font-size: 0.95rem;
    }
    nav button { background: var(--white); color: var(--green); }
    button:focus-visible, input:focus-visible, label:focus-visible {
      outline: 3px solid #f0bd4b;
      outline-offset: 3px;
    }
    .page { display: none; padding: 24px 16px; }
    .page.active { display: block; }
    .container { max-width: 1000px; margin: 0 auto; }
    .hero {
      max-width: 1000px;
      margin: 0 auto;
      padding: 46px 22px;
      min-height: 340px;
      border-radius: 22px;
      color: var(--white);
      text-align: center;
      background: linear-gradient(135deg, #07552f, #0b9b59);
      display: flex;
      flex-direction: column;
      align-items: center;
      justify-content: center;
    }
    .hero .leaf { font-size: 4rem; }
    .hero h2 { font-size: clamp(2rem, 5vw, 3rem); margin: 8px 0; }
    .hero p { max-width: 720px; margin-top: 10px; }
    .btn { margin: 12px 5px 0; }
    .btn.green { background: var(--green); color: var(--white); }
    .btn.white { background: var(--white); color: var(--green); }
```

Complete Source Code (continued)

```
    .section-title { margin: 28px 0 14px; }
    .cards { display: grid; grid-template-columns: repeat(3, minmax(0, 1fr)); gap: 15px; }
    .card, .stat {
      background: var(--white);
      padding: 20px;
      border: 1px solid #00000008;
      border-radius: 16px;
      box-shadow: 0 5px 18px #0000000c;
      margin: 12px 0;
    }
    .card h3 { margin-bottom: 8px; color: var(--green); }
    .stats h2 { color: var(--green); font-size: 2rem; }
    .stat { text-align: center; }
    .upload {
      border: 2px dashed #58ae7d;
      padding: 25px;
      text-align: center;
      border-radius: 16px;
      background: #f8fffa;
    }
    input[type="file"] { display: block; max-width: 100%; margin: 12px auto; }
    #preview, #resultImage {
      display: none;
      max-width: 100%;
      max-height: 300px;
      object-fit: contain;
      margin: 18px auto 0;
      border-radius: 14px;
    }
    .result { display: none; }
    .badge {
      display: inline-block;
      background: var(--mint);
      color: var(--green);
      border-radius: 20px;
      padding: 4px 10px;
      font-weight: bold;
      margin: 6px 0 12px;
    }
    .progress { height: 10px; background: #e2e8e4; border-radius: 10px; overflow: hidden; margin: 8px 0
    16px; }
    .progress div { width: 94%; height: 100%; background: var(--green); }
    .solution {
      display: flex;
      gap: 12px;
      padding: 14px;
      margin: 10px 0;
      background: #effaf3;
      border-radius: 12px;
    }
    .number {
      flex: 0 0 32px;
      width: 32px;
      height: 32px;
      display: grid;
      place-items: center;
      background: var(--green);
      color: white;
      border-radius: 50%;
      font-weight: bold;
    }
    .warning { background: var(--warning); padding: 15px; border-radius: 12px; margin-top: 16px; }
    footer { background: var(--dark); color: white; text-align: center; padding: 22px 15px; margin-top:
    20px; }
    .muted { color: var(--muted); font-size: 0.93rem; }
    @media (max-width: 700px) {
      .cards { grid-template-columns: 1fr; }
      .page { padding: 18px 12px; }
      .hero { padding: 34px 16px; }
      nav button { font-size: 0.85rem; padding: 8px 10px; }
    }
  </style>
</head>
<body>
<header>
```

Complete Source Code (continued)

```
  <h1>PlantCare AI</h1>
  <nav aria-label="Main navigation">
    <button type="button" onclick="showPage('home')">Home</button>
    <button type="button" onclick="showPage('detect')">AI Detection</button>
    <button type="button" onclick="showPage('solution')">Solution</button>
    <button type="button" onclick="showPage('prevention')">Prevention</button>
    <button type="button" onclick="showPage('about')">About</button>
  </nav>
</header>

<main>
  <section id="home" class="page active">
    <div class="hero">
      <div class="leaf" aria-hidden="true">■</div>
      <h2>PlantCare AI</h2>
      <h3>Plant Disease Detection and Care Guidance</h3>
      <p>Upload a plant leaf photo, preview the image, and explore sample plant-care guidance for a
    smart-farming project demonstration.</p>
      <div>
        <button class="btn white" type="button" onclick="showPage('detect')">Upload Leaf Image</button>
        <button class="btn white" type="button" onclick="openCamera()">Take Photo</button>
      </div>
    </div>
    <div class="container">
      <h2 class="section-title">Smart Farming with AI</h2>
      <div class="cards">
        <article class="card"><h3>Easy Image Selection</h3><p>Select a JPG, PNG, or supported camera
    image and see its preview.</p></article>
        <article class="card"><h3>Detection Demo</h3><p>Run a fixed sample result to demonstrate how a
    future AI detection workflow could look.</p></article>
        <article class="card"><h3>Care Guidance</h3><p>Review practical prevention ideas and five sample
    steps for tomato early blight management.</p></article>
      </div>
      <h2 class="section-title">Dashboard</h2>
      <div class="cards stats">
        <div class="stat"><h2 id="total">0</h2><p>Total Scans</p></div>
        <div class="stat"><h2 id="healthy">0</h2><p>Healthy Results</p></div>
        <div class="stat"><h2 id="diseased">0</h2><p>Sample Disease Results</p></div>
      </div>
    </div>
  </section>

  <section id="detect" class="page">
    <div class="container">
      <h2>Plant Leaf Image Demo</h2>
      <p class="muted">Choose an image or use the camera option on a supported mobile device.</p>
      <div class="card">
        <div class="upload">
          <h3>Select a Plant Leaf Image</h3>
          <p>Supported image types: JPG, PNG, and other browser-supported image formats.</p>
          <input type="file" id="leafInput" accept="image/*" aria-label="Choose a plant leaf image">
          <button class="btn green" type="button" onclick="openCamera()">Take Photo</button>
          <img id="preview" alt="Preview of the selected plant leaf">
        </div>
        <p id="fileStatus" class="muted" role="status" aria-live="polite">No image selected.</p>
        <button class="btn green" type="button" onclick="detectDisease()">Run Demo Detection</button>
        <div class="warning"><strong>Demo limitation:</strong> This version does not run a trained AI
    model. The displayed disease and confidence are fixed sample data for a college project
    demonstration.</div>
      </div>
      <div class="card result" id="result" aria-live="polite">
        <h2>Sample Detection Result</h2>
        <img id="resultImage" alt="Selected plant leaf used in the demonstration">
        <p><strong>Sample plant:</strong> Tomato</p>
        <p><strong>Sample result:</strong> Early Blight</p>
        <p><strong>Demo confidence value:</strong> 94% (fixed sample; not calculated from the uploaded
    image)</p>
        <div class="progress" aria-label="Illustrative sample confidence"><div></div></div>
        <p><strong>Sample severity:</strong> Moderate</p>
        <p><strong>Common symptoms:</strong> Dark spots on older leaves, yellowing around spots, and
    gradual leaf damage.</p>
        <button class="btn green" type="button" onclick="showPage('solution')">View Care Steps</button>
      </div>
    </div>
  </section>
```

Complete Source Code (continued)

```
  <section id="solution" class="page">
    <div class="container">
      <h2>Disease Management: Sample Tomato Early Blight</h2>
      <p class="muted">These are general educational suggestions, not a diagnosis or a substitute for
    local agricultural advice.</p>
      <div class="card">
        <div class="solution"><span class="number">1</span><div><strong>Remove affected
    leaves.</strong><p>Remove badly affected plant material carefully and dispose of it according to
    local guidance.</p></div></div>
        <div class="solution"><span class="number">2</span><div><strong>Provide suitable plant
    spacing.</strong><p>Good airflow can help reduce prolonged moisture around leaves.</p></div></div>
        <div class="solution"><span class="number">3</span><div><strong>Water near the
    soil.</strong><p>Avoid unnecessary wetting of leaves and water according to the crop's
    needs.</p></div></div>
        <div class="solution"><span class="number">4</span><div><strong>Keep the growing area
    clean.</strong><p>Remove fallen plant debris and monitor the crop regularly.</p></div></div>
        <div class="solution"><span class="number">5</span><div><strong>Ask an agriculture professional
    about treatment.</strong><p>Use only locally approved products as directed on their labels and by
    qualified guidance.</p></div></div>
      </div>
    </div>
  </section>

  <section id="prevention" class="page">
    <div class="container">
      <h2>Prevention and Plant Care</h2>
      <div class="cards">
        <article class="card"><h3>Regular Inspection</h3><p>Inspect leaves and stems regularly for
    unusual spots, yellowing, wilting, or changes in growth.</p></article>
        <article class="card"><h3>Water Management</h3><p>Adjust watering to the crop, soil, weather, and
     growth stage. Avoid unnecessary overwatering.</p></article>
        <article class="card"><h3>Balanced Nutrition</h3><p>Choose fertilizer based on crop requirements
    and soil advice. Avoid excessive fertilizer application.</p></article>
        <article class="card"><h3>Healthy Planting Material</h3><p>Use healthy seeds or seedlings and
    follow recommended crop hygiene practices.</p></article>
        <article class="card"><h3>Good Airflow</h3><p>Follow suitable plant spacing and manage weeds
    where appropriate.</p></article>
        <article class="card"><h3>Expert Support</h3><p>Contact a qualified agriculture professional if
    symptoms spread quickly or the cause is uncertain.</p></article>
      </div>
      <div class="warning"><strong>Important:</strong> This website is a college project prototype.
    Confirm serious or unfamiliar plant problems with a qualified agriculture professional.</div>
    </div>
  </section>

  <section id="about" class="page">
    <div class="container">
      <h2>About PlantCare AI</h2>
      <div class="card"><h3>Problem</h3><p>Farmers may find it difficult to notice and identify plant
    diseases early. Manual inspection can take time, especially across large fields.</p></div>
      <div class="card"><h3>Proposed Solution</h3><p>PlantCare AI demonstrates an interface where a user
    selects a leaf image and views a sample result alongside plant-care guidance. The current version is
    a front-end prototype and does not classify images using a trained model.</p></div>
      <div class="card"><h3>Illustrative Sample Categories</h3><p>Tomato — Early Blight; Rice — Leaf
    Blast; Potato — Late Blight; Chili — Leaf Curl; Cotton — Leaf Spot. These are examples only and are
    not detected by this prototype.</p></div>
      <div class="card"><h3>Technology Used</h3><p>HTML structures the pages, CSS styles the responsive
    interface, and JavaScript manages navigation, image preview, demo results, and dashboard
    counters.</p></div>
      <div class="card"><h3>Future Scope</h3><p>Connect a trained TensorFlow/Keras plant disease model
    through a backend API, add validated crop-specific guidance, support regional languages and voice
    input, and store scan history securely.</p></div>
    </div>
  </section>
</main>

<footer>
  <strong>PlantCare AI</strong><br>
  Smart Plant Care | College AI Immersion Project
</footer>

<script>
  let totalScans = 0;
  let healthyResults = 0;
```

Complete Source Code (continued)

```
  let diseaseResults = 0;

  const leafInput = document.getElementById("leafInput");
  const preview = document.getElementById("preview");
  const resultImage = document.getElementById("resultImage");
  const result = document.getElementById("result");
  const fileStatus = document.getElementById("fileStatus");

  function showPage(pageId) {
    document.querySelectorAll(".page").forEach(function (page) {
      page.classList.remove("active");
    });
    const selectedPage = document.getElementById(pageId);
    if (selectedPage) selectedPage.classList.add("active");
    window.scrollTo({ top: 0, behavior: "smooth" });
  }

  leafInput.addEventListener("change", function () {
    const file = leafInput.files && leafInput.files[0];
    result.style.display = "none";

    if (!file) {
      preview.style.display = "none";
      fileStatus.textContent = "No image selected.";
      return;
    }

    if (!file.type.startsWith("image/")) {
      leafInput.value = "";
      preview.style.display = "none";
      fileStatus.textContent = "Please select a valid image file.";
      alert("Please select a valid image file.");
      return;
    }

    const imageUrl = URL.createObjectURL(file);
    preview.src = imageUrl;
    resultImage.src = imageUrl;
    preview.style.display = "block";
    fileStatus.textContent = "Selected image: " + file.name;
  });

  function openCamera() {
    showPage("detect");
    setTimeout(function () {
      leafInput.click();
    }, 150);
  }

  function detectDisease() {
    if (!leafInput.files || !leafInput.files[0]) {
      alert("Please upload or take a plant leaf photo first.");
      return;
    }

    totalScans += 1;
    diseaseResults += 1;

    document.getElementById("total").textContent = totalScans;
    document.getElementById("healthy").textContent = healthyResults;
    document.getElementById("diseased").textContent = diseaseResults;

    result.style.display = "block";
    result.scrollIntoView({ behavior: "smooth", block: "start" });
  }
</script>
</body>
</html>
```